

概率统计 20-21-2(A)标准答案及评分标准

一、选择题

1)C 2) B 3)B, 4) B, 5) B

二、填空题

1) 0.3;

2)0.0486

3) 180

4)0.1587

5)2.65

6) 14

7) 2

8) 27

$$9) F(x) = \begin{cases} 0 & x < 1 \\ 0.3 & 1 \leq x < 2 \\ 0.8 & 2 \leq x < 3 \\ 1 & x \geq 3 \end{cases}$$

$$10) f_Y(y) = \begin{cases} \frac{3}{32}(y^2 + 2y - 3) & -5 < y < -3 \\ 0 & \text{其它} \end{cases}; \begin{cases} 0.09375(y^2 + 2y - 3) & -5 < y < -3 \\ 0 & \text{其它} \end{cases}$$

11) 8

12) 大于 0.95

13) 2.4

三、 (1) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1; \dots\dots\dots 2'$

$$a \int_0^1 \int_0^{2-2x} xy dy dx = 1; \dots\dots\dots 2'$$

$$a = 6. \dots\dots\dots 1'$$

$$(2) f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy \dots\dots\dots 2'$$

$$\text{当 } 0 < x < 1 \text{ 时 } f_X(x) = \int_0^{2-2x} axy dy = 12(1-x)^2 \dots\dots\dots 2'$$

$$\text{当 } x \leq 0, \text{ 或 } x \geq 1 \text{ 时 } f_X(x) = 0 \dots\dots\dots 1'$$

$$(3) f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} = \begin{cases} \frac{y}{2(1-x)^2} & 0 < y < 2(1-x) \\ 0 & \text{其它} \end{cases} \quad (0 < x < 1) \dots\dots\dots 2'$$

$$f_{Y|X}(y|0.5) = \begin{cases} 2y & 0 < y < 1 \\ 0 & \text{其它} \end{cases} \dots\dots\dots 2'$$

$$P(Y < 0.5 | X = 0.5) = \int_{-\infty}^{0.5} f_{Y|X}(y|0.5) dy = \int_0^{0.5} 2y dy = 0.25 \dots\dots\dots 1'$$

四、A1, A2, A3 分别表示 产品由甲、乙、丙厂家生产;

B 表示抽到两件均为次品. 则

$$P(A_1) = 0.6; P(A_2) = 0.2; P(A_3) = 0.2; \dots\dots\dots 1'$$

$$P(B|A_1) = 0.05^2; P(B|A_2) = 0.1^2; P(B|A_3) = 0.1^2; \dots\dots\dots 1'$$

$$(1) P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3) \dots\dots\dots 2'$$

$$= 0.6 * 0.05^2 + 0.2 * 0.1^2 + 0.2 * 0.1^2 = 0.0055 \dots\dots\dots 2'$$

(2)

$$P(A_3|B) = \frac{P(A_3B)}{P(B)} = \frac{P(A_3)P(B|A_3)}{P(B)} \dots\dots\dots 2'$$

$$= \frac{0.2 * 0.1^2}{0.0055} = \frac{4}{11} \approx 0.364 \dots\dots\dots 2'$$

五、X和Y的概率密度为:

$$f_X(x) = \begin{cases} e^{-x} & x > 0 \\ 0 & \text{其它} \end{cases}, f_Y(y) = \begin{cases} 1 & 1 < y < 2 \\ 0 & \text{其它} \end{cases} \dots\dots\dots 2'$$

X和Y的联合密度为:

$$f(x, y) = \begin{cases} e^{-x} & x > 0, 1 < y < 2 \\ 0 & \text{其它} \end{cases} \dots\dots\dots 1'$$

Z的分布函数 $F_Z(z) = P(Z \leq z) = P(X + Y \leq z) \dots\dots\dots 1'$

当 $z < 1$ 时, $F_Z(z) = 0$;

$$\text{当 } 1 \leq z \leq 2 \text{ 时, } F_Z(z) = \iint_{x+y \leq z} f(x, y) dx dy \dots\dots\dots 2'$$

$$= \int_1^z \int_0^{z-y} e^{-x} dx dy \dots\dots\dots 1'$$

$$= z + e^{1-z} - 2 \dots\dots\dots 1'$$

当 $z > 2$ 时

$$F_Z(z) = \iint_{x+y \leq z} f(x, y) dx dy = \int_1^2 \int_0^{z-y} e^{-x} dx dy = 1 - e^{-z}(e^2 - e) \dots\dots\dots 1'$$

Z的概率密度为

$$f_Z(z) = [F_Z(z)]' = \begin{cases} 1 - e^{1-z} & 1 < z < 2 \\ e^{-z}(e^2 - e) & z \geq 2 \\ 0 & z < 1 \end{cases} \dots\dots\dots 1'$$

或者:

X和Y的概率密度为:

$$f_X(x) = \begin{cases} e^{-x} & x > 0 \\ 0 & \text{其它} \end{cases}, f_Y(y) = \begin{cases} 1 & 1 < y < 2 \\ 0 & \text{其它} \end{cases} \dots\dots\dots 2'$$

$$f_Z(z) = \int_{-\infty}^{+\infty} f_X(x)f_Y(z-x)dx \dots\dots\dots 2'$$

当 $1 < z < 2$ 时

$$f_Z(z) = \int_0^{z-1} e^{-x} dx \dots\dots\dots 2'$$

$$= 1 - e^{-(z-1)} \dots\dots\dots 1'$$

$$\text{当 } z \geq 2 \text{ 时, } f_z(z) = \int_{z-2}^{z-1} e^{-x} dx = e^{-z}(e^2 - e) \dots\dots\dots 2'$$

$$\text{当 } z \leq 1, f_z(z) = 0 \dots\dots\dots 1'$$

六、 X_i 表示第*i*次出现的点数; $P(X_i = k) = \frac{1}{6}, k = 1, 2, 3, 4, 5, 6; \dots\dots\dots 2$

$$\mu = EX_i = 3.5; \sigma^2 = DX_i = \frac{35}{12}; n=100; \dots\dots\dots 2$$

所求改为为:

$$P\left(\sum_{i=1}^n X_i \leq 370\right) \approx \Phi\left(\frac{370 - n\mu}{\sqrt{n} \sigma}\right) \dots\dots\dots 4'$$

$$= \Phi\left(\frac{2\sqrt{12}}{\sqrt{35}}\right) = \Phi(1.171) \dots\dots\dots 1'$$

$$\text{七、(1) 似然函数为: } L(\theta) = \prod_{i=1}^n f(X_i, \theta) = \prod_{i=1}^n \theta^{-\frac{X_i-2}{3}} (1-\theta)^{\frac{X_i+1}{3}}$$

$$= \theta^{-\sum \frac{X_i-2}{3}} (1-\theta)^{\sum \frac{X_i+1}{3}} \dots\dots\dots 2'$$

$$\ln L(\theta) = -\sum \frac{X_i-2}{3} \ln \theta + \sum \frac{X_i+1}{3} \ln(1-\theta) \dots\dots\dots 1'$$

$$[\ln L(\theta)]' = -\frac{1}{\theta} \sum \frac{X_i-2}{3} - \frac{1}{1-\theta} \sum \frac{X_i+1}{3} = 0 \dots\dots 2'$$

$$\hat{\theta} = \frac{2}{3} - \frac{1}{3} \bar{X}; \bar{X} = \frac{1}{n} \sum X_i \dots\dots 1'$$

$$(2) \quad E\hat{\theta} = E\left(\frac{2}{3} - \frac{1}{3} \bar{X}\right) = \frac{2}{3} - \frac{1}{3} EX_1 \dots\dots\dots 1'$$

$$EX_1 = (-1)\theta + 2(1-\theta) = 2 - 3\theta \dots\dots\dots 1'$$

$$E\hat{\theta} = \frac{2}{3} - \frac{1}{3} (2 - 3\theta) = \theta \dots\dots\dots 1'$$

$$\hat{\theta} \text{ 是 } \theta \text{ 的无偏估计} \dots\dots\dots 1'$$

八、(1) $n = 25, \alpha = 0.05,$

$$\text{检验统计量 } T = \frac{\bar{X} + 4}{S_n} \sqrt{n} | H_0 \sim t(n-1)$$

$$\text{拒绝域: } D = \{T < -t_\alpha(n-1)\} = \{T < -1.7113\} \dots\dots\dots 4'$$

$$\bar{x} = -5, s_n = 2$$

$$T \text{ 的观测值: } T = \frac{-5 + 4}{2} \sqrt{25} = -2.5$$

$$-2.5 < -1.7113, \dots\dots\dots 1'$$

所以, 拒绝原假设。 $\dots\dots\dots 1'$

$$(2) \sigma^2 \text{ 的置信度为 } 95\% \text{ 的置信区间为: } \left[\frac{(n-1)S_n^2}{\chi_{0.025}^2(24)}, \frac{(n-1)S_n^2}{\chi_{0.975}^2(24)} \right] \dots\dots 2'$$

$$= \left[\frac{24 \times 2^2}{39.36}, \frac{24 \times 2^2}{12.4} \right] = [2.439, 7.742] \dots\dots\dots 2'$$